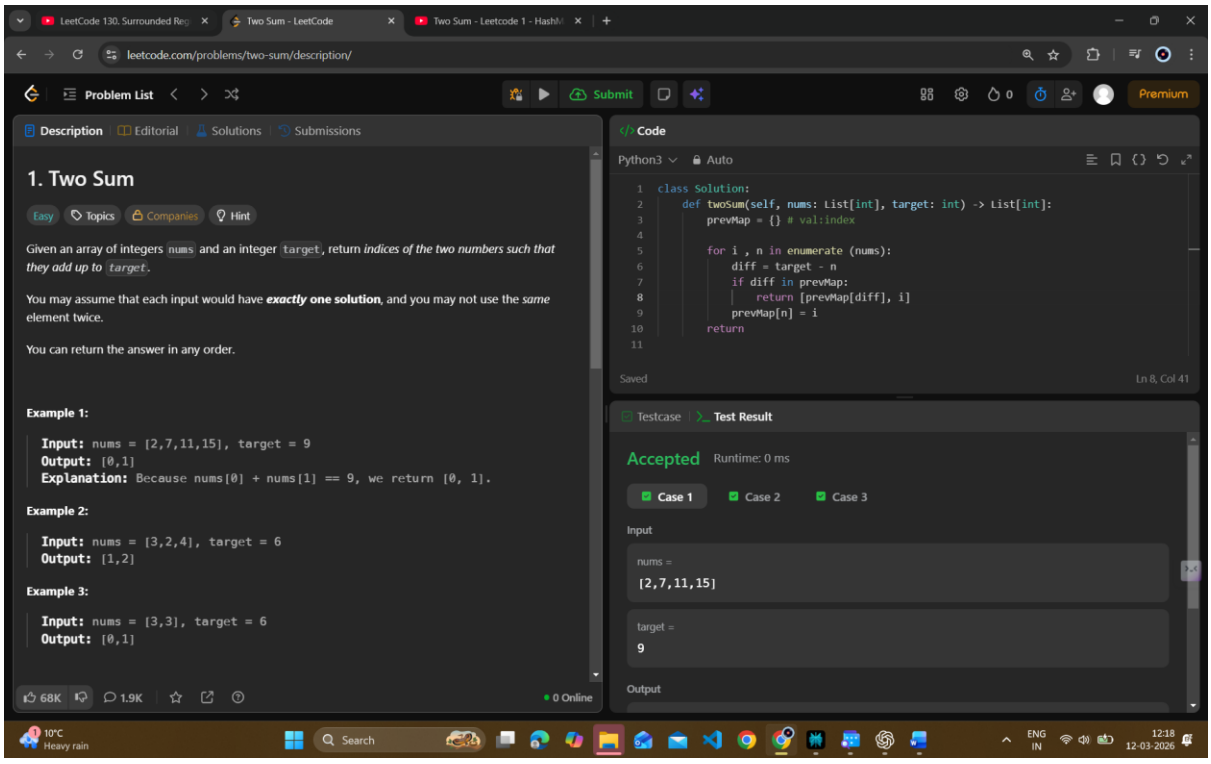
1. Two Sum

Given an array of integers nums and an integer target, return *indices of the two numbers such that they add up to target*.
You may assume that each input would have ***exactly* one solution**, and you may not use the *same* element twice.
You can return the answer in any order.

Example 1:
Input: nums = [2,7,11,15], target = 9
Output: [0,1]
Explanation: Because nums[0] + nums[1] == 9, we return [0, 1].
Example 2:
Input: nums = [3,2,4], target = 6
Output: [1,2]
Example 3:
Input: nums = [3,3], target = 6
Output: [0,1]

Constraints:
- $2 <= nums.length <= 10^4$
- $-10^9 <= nums[i] <= 10^9$
- $-10^9 <= target <= 10^9$
- **Only one valid answer exists.**

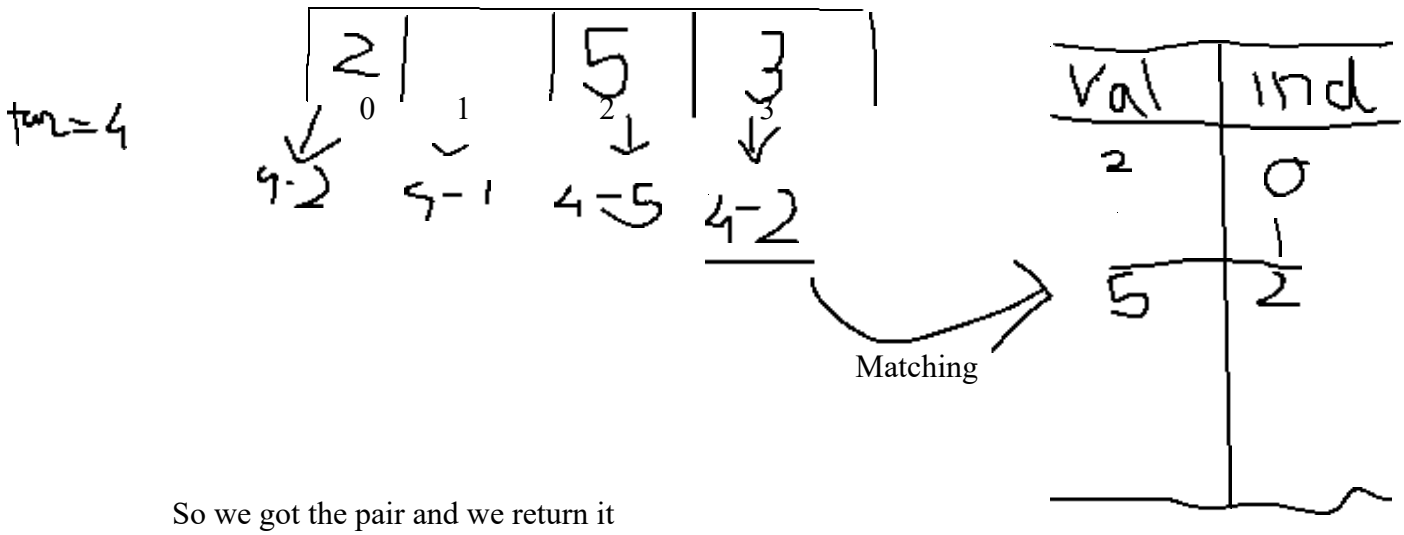


Solution:

Approch 1 To understand take an array and apply bruite force first so lets consider array [2,7,11,15] target = 9

So compare each element with each element to n element takes n traverse so the thing is this one is having the complexity as $O(n^2)$

Approch 2 : we have to find a optimal way for this so we can use a hashmap as we all know it stores the values and indexes
So let us provide a hashmap



So we got the pair and we return it

So lets explain the code

So as we can see we have given

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class Solution:
    def twoSum(self, nums: List[int], target: int) -> List[int]:
        # so this is given to us and we have to perform our operations so a list is created and a target is set to so first is create a hashmap
        prevMap = {} # val: index    so this is our map

        for i , n in enumerate(nums):
            diff = target -n
            if diff in prevMap:
                return [prevMap[diff], i]
            prevMap[n]=i
        return
```